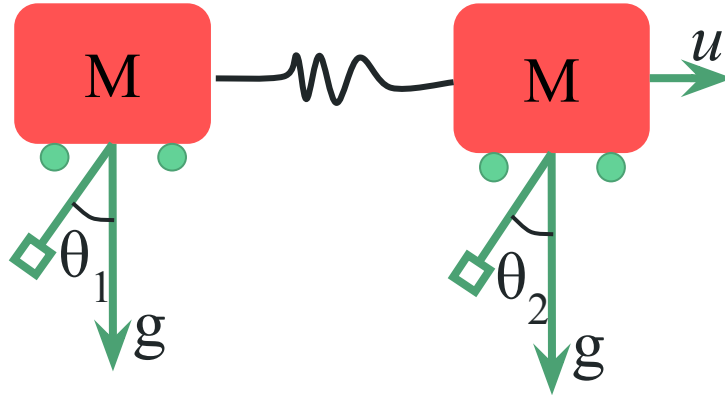


Inverted Pendulum: Swing-up, Balance, Catching

Raaghav Thirumaligai, Cole Giusto, Tien Nguyen
June 2, 2025

Heartbreaking backstory

- The vibrational stabilization
- The double cart pendula spring system

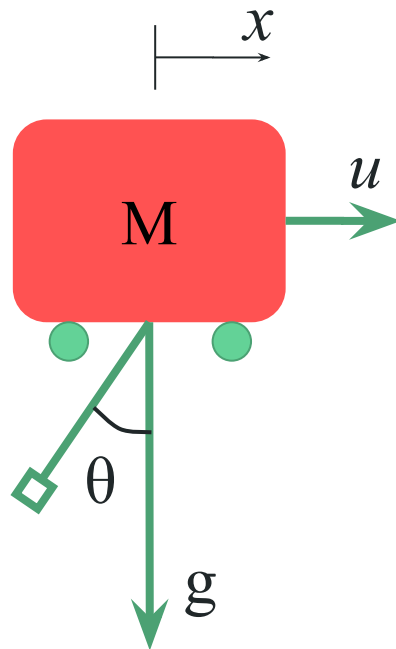


System Description

- Asymmetric metal rod with end pinned to cart
- Encoder for angle of rod
- Single Motor to drive cart
- Encoder for position of cart

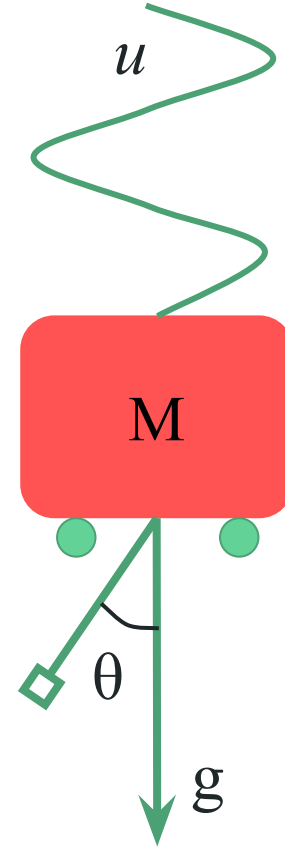
System Objectives

- Stabilize downward equilibrium (for faster catching)
- Stabilize upward equilibrium
- State transition between equilibria

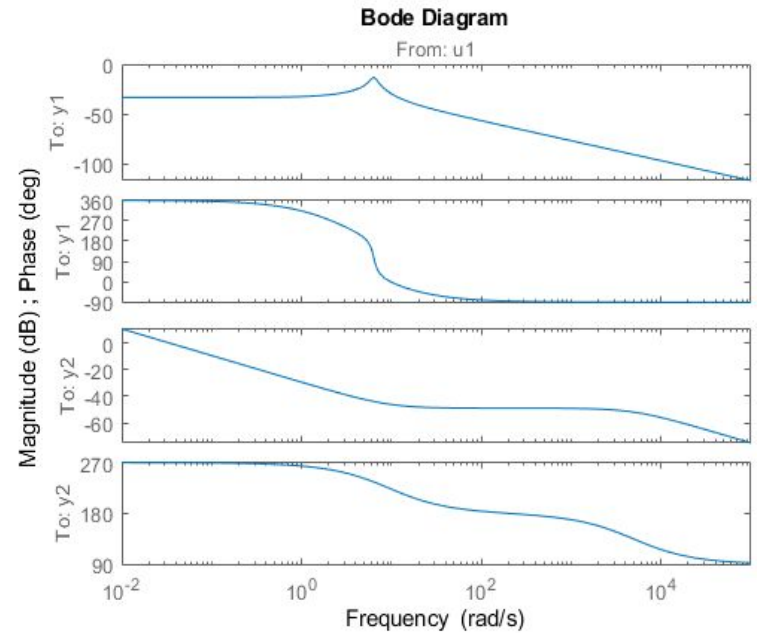
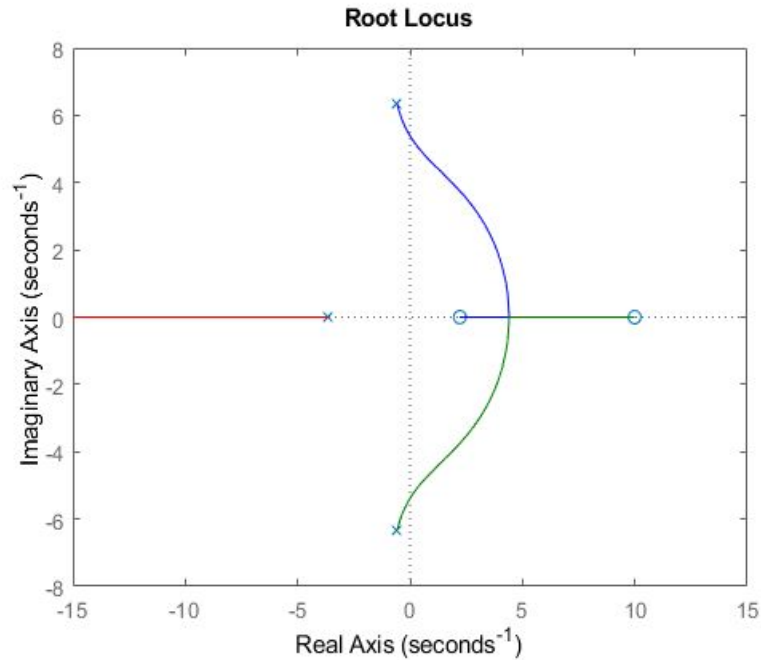


Model Identification

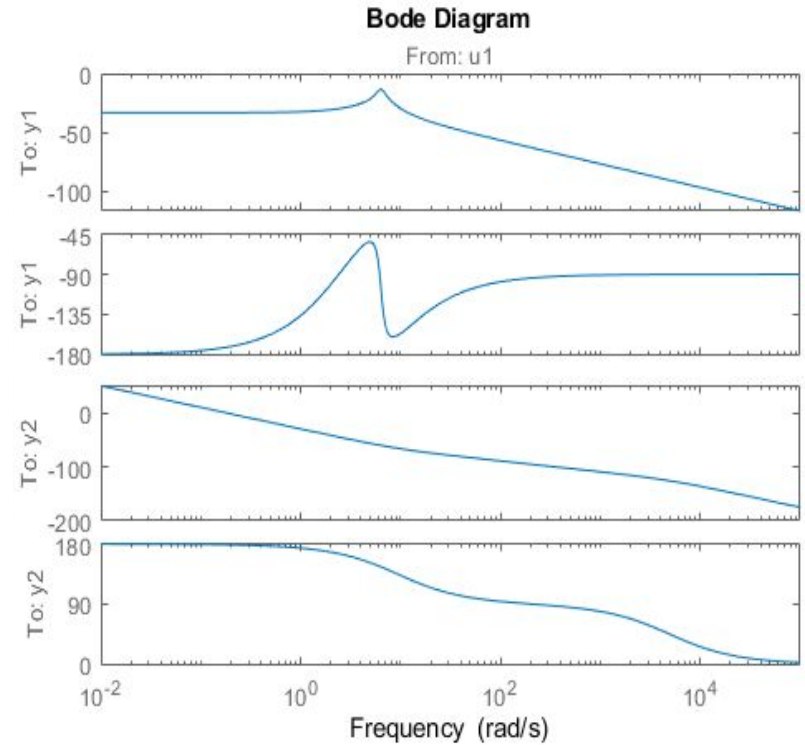
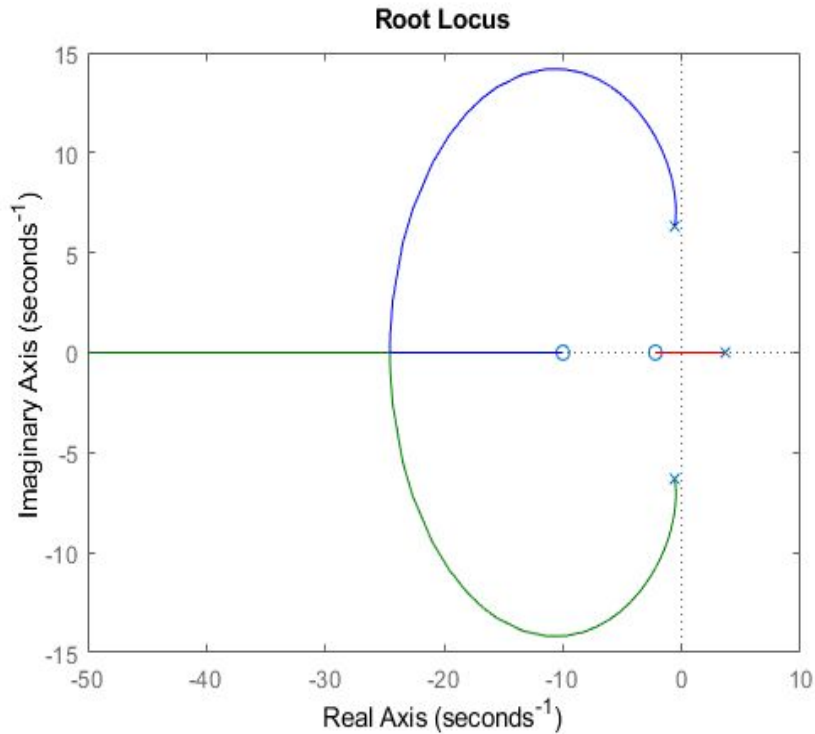
- Logarithmic Sine sweep ($f \in [0.3, 3]$ Hz)
- Tfest on 30 experiments to estimate linear model
- Assumed integrator for cart, fit Voltage to Velocity model
- Used 2 poles, 1 zero for cart, 3 poles 2 zeros for pendulum



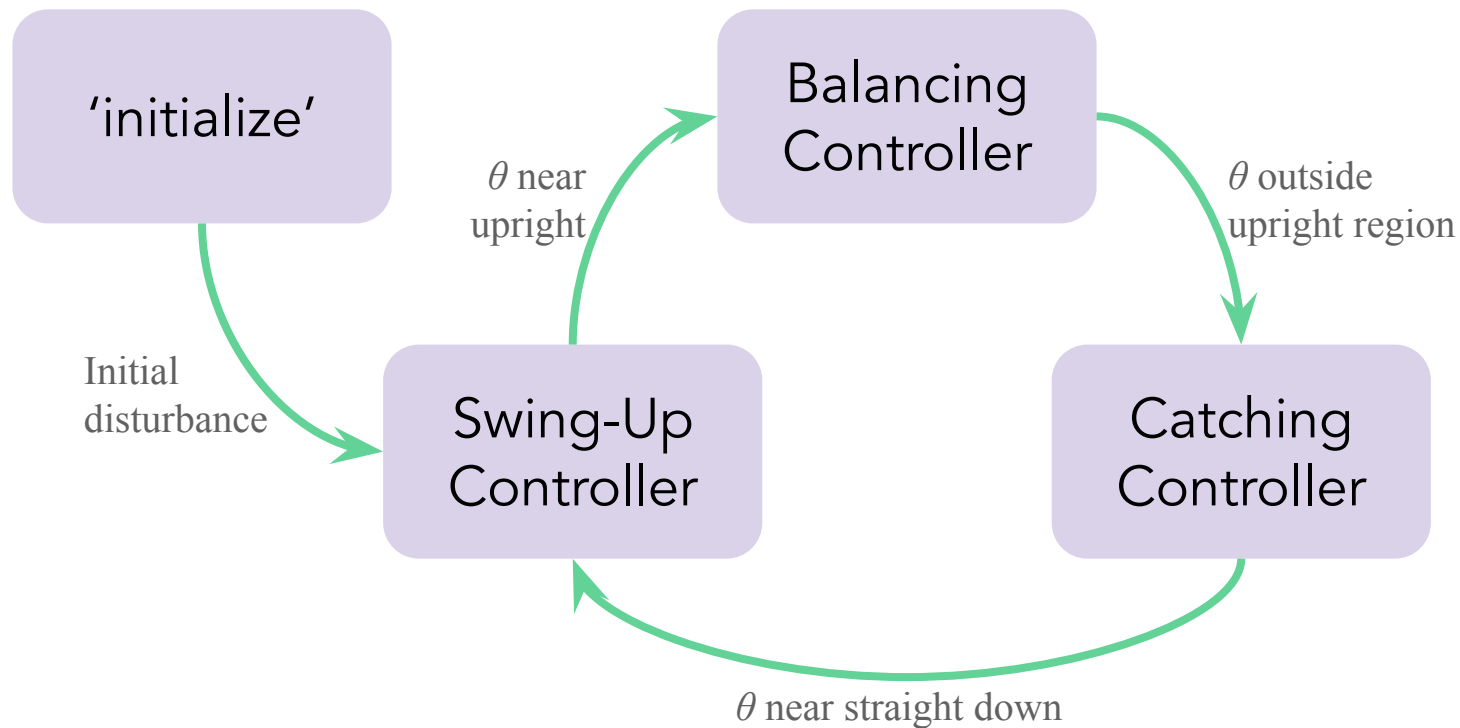
Identification of the stable equilibrium



Modify for unstable equilibrium

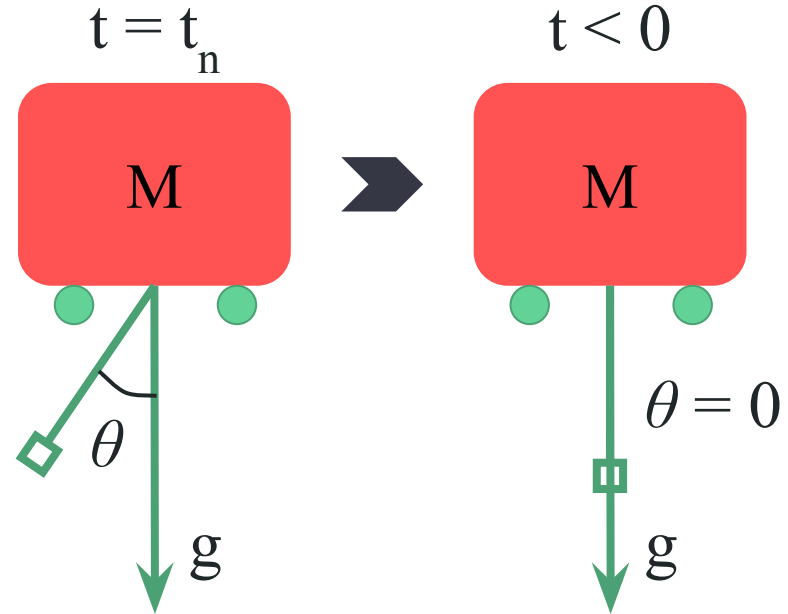


State Machine



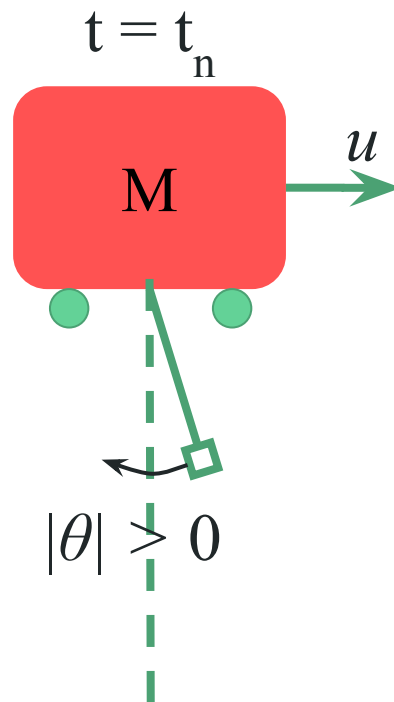
Initialization

- Accurate initial theta with gravity aligning the pendulum
- Important to have accurate reading for inverted upward position



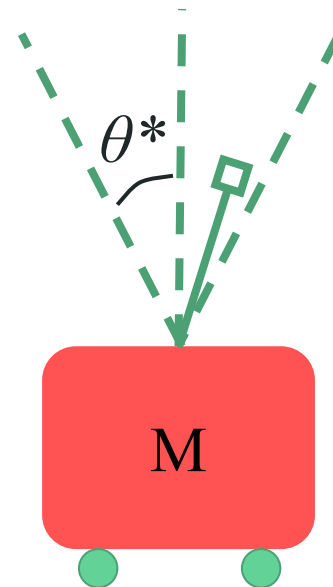
Swing Up Controller

- Energy addition method:
 - Around every zero crossing of $\dot{\theta}$, the cart moves in direction opposite to the direction of the velocity to increase the energy in the pendulum
 - Continues until pendulum is able to fully “swing up”
 - Simple proportional controller regulates cart position
- The activation is limited to a range $\theta \in [-30, 30]^\circ$ to try and reach inverted state with a small velocity



Balancing Controller

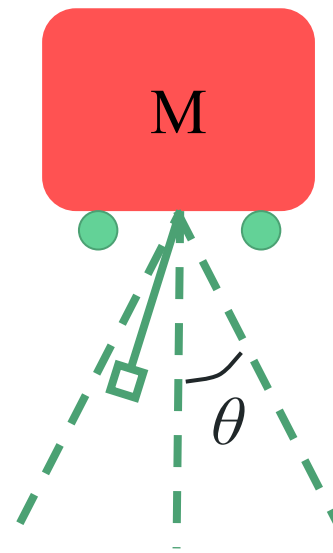
- Activates upon state entering:
 - $\theta^* \in [-23, 23]^\circ$
 - Swing Up getting 'close'
- Stabilizes unstable equilibrium
- Utilizes LQR/LQG with the model identified earlier to regulate both cart position and pendulum angle
- Carefully tuned parameters to be as robust as possible



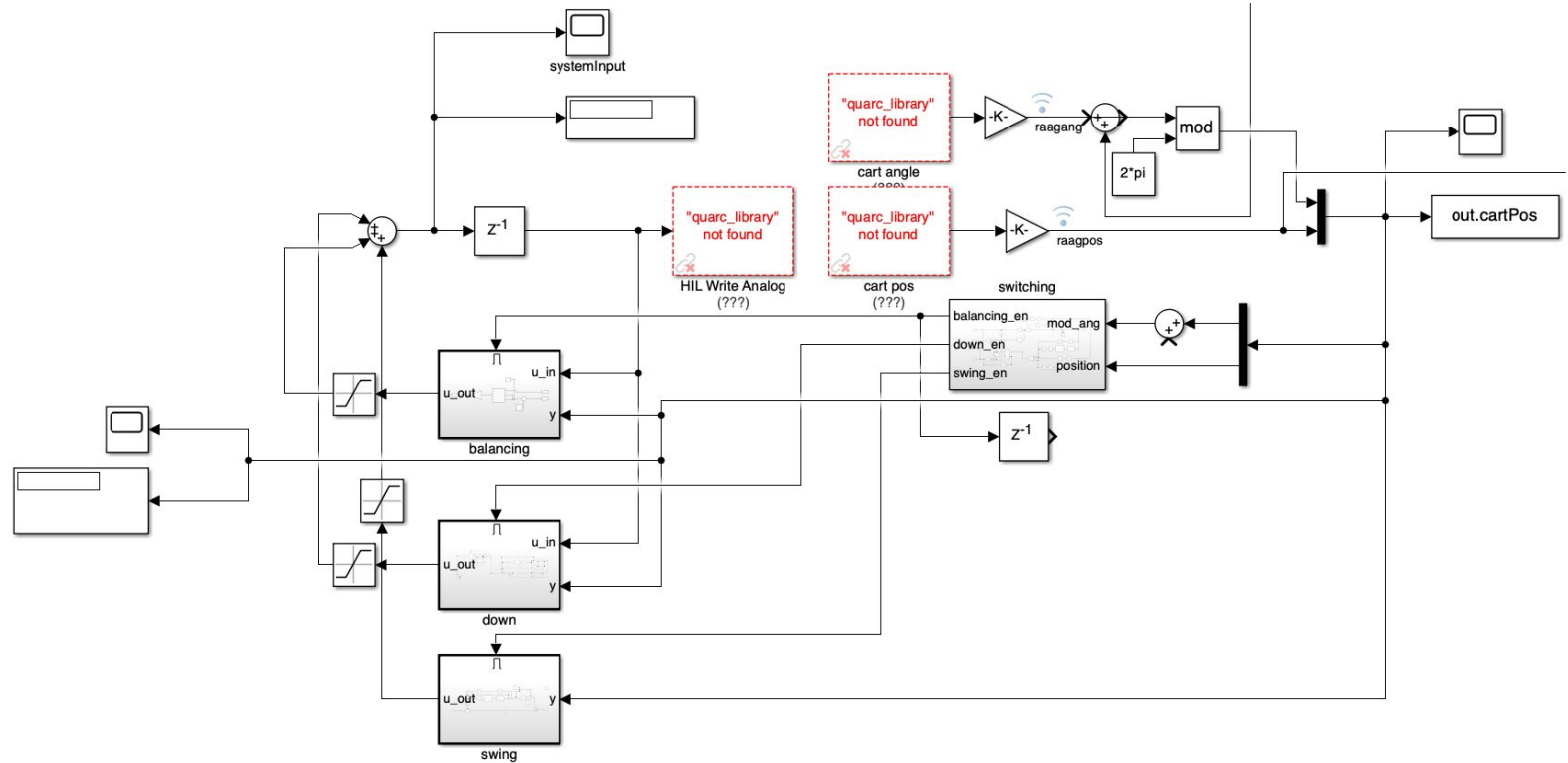
Note: $\theta^ = \theta \bmod(2\pi) - \pi$*

Catching Controller

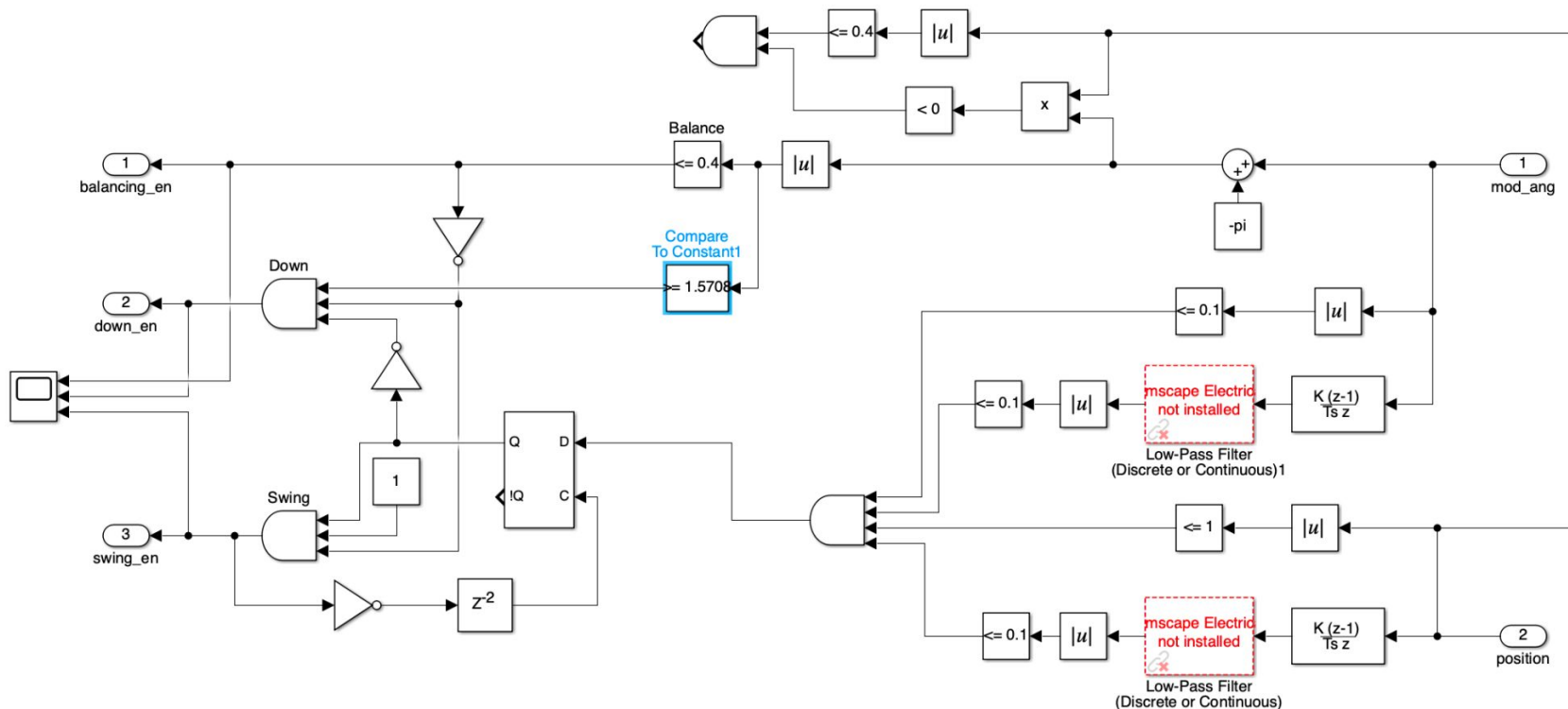
- Activates upon state entering:
 - $\theta \in [-90, 90]^\circ$
 - Pendulum Falls
- Also utilizes LQR/LQG, this time to return to the bottom stable position
- Brings pendulum back into predictable state to start Swing Up Controller



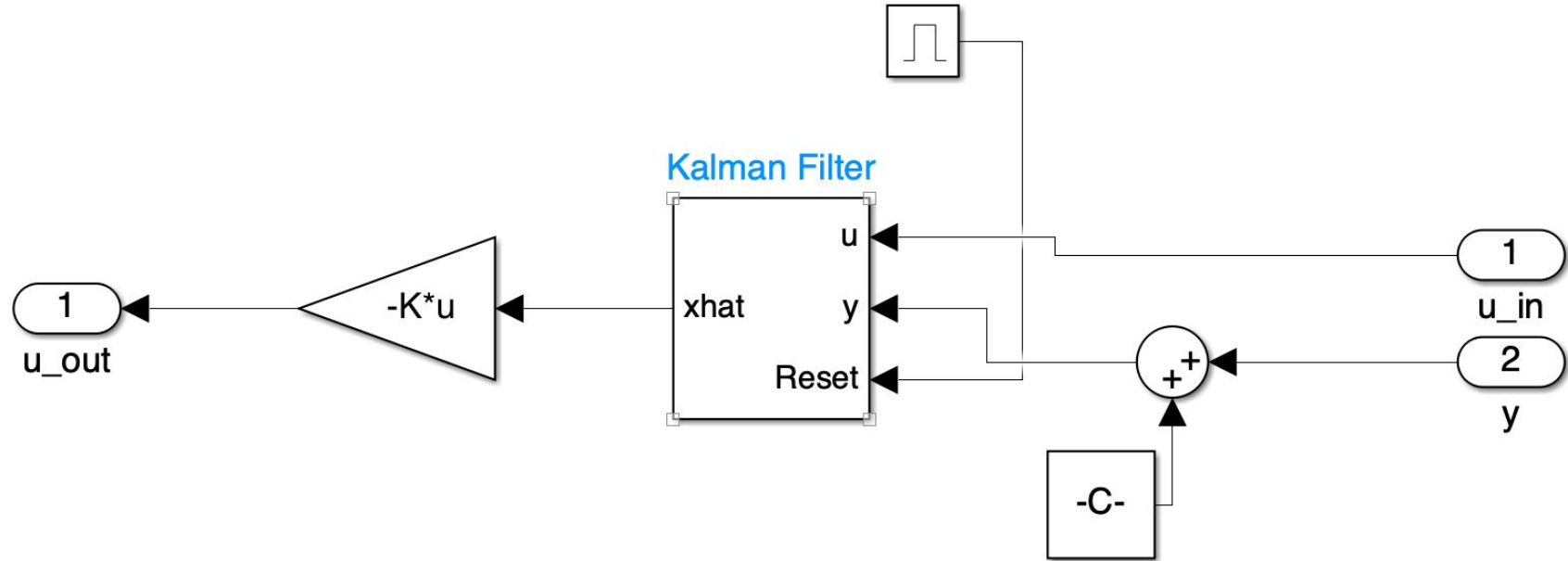
Full Controller



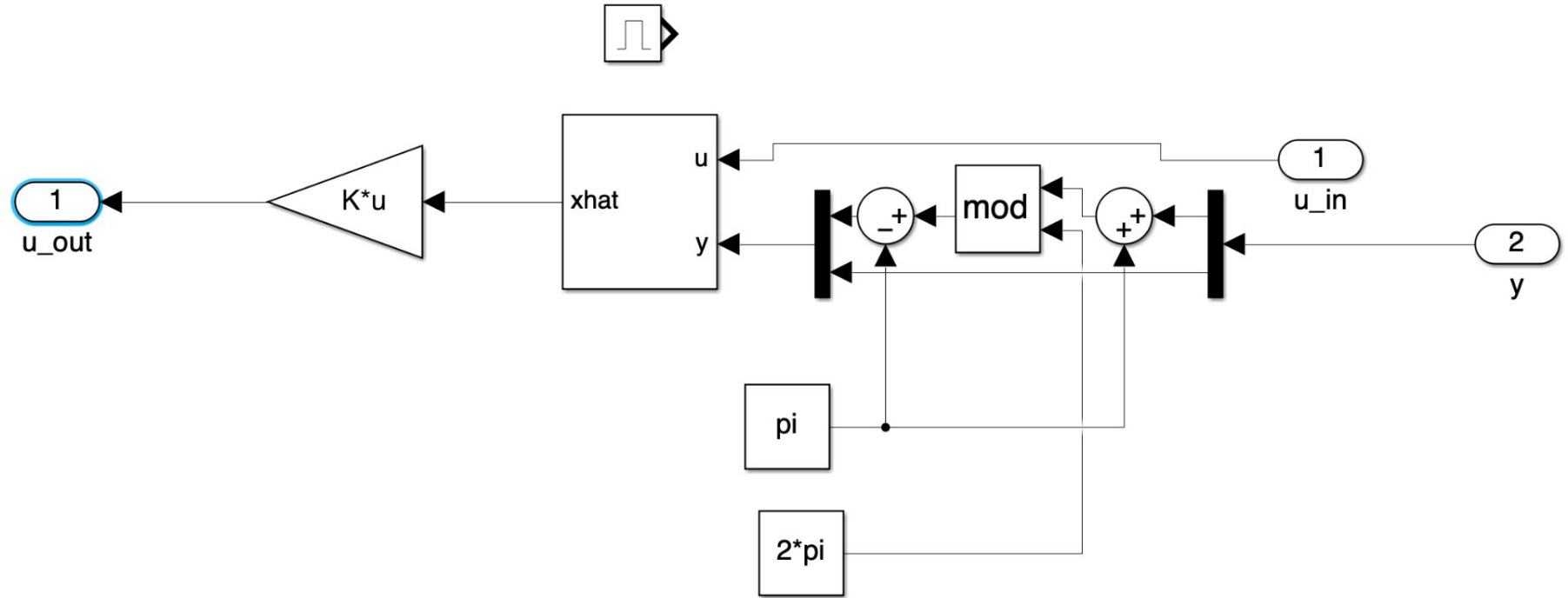
Switching



Balancing



Catching



Demo:

Go to Lab Station

Future Work

- Introduce correction term to account for steady state error
- More robust failure logic

Acknowledgement

Zeki

Joao